



WhatsApp Bot Operations Manual (Field Engineering)

Standard operating procedures for on-site field engineers conducting mobile tool checkouts, status queries, and custody transfers.

PART 2: WHATSAPP BOT OPERATIONS (FIELD ENGINEERING)

Field engineers operate in remote mobile environments without requiring VPN installations or laptop access. The official **Entelso WhatsApp Bot** serves as an intelligent mobile assistant, enabling engineers to verify equipment safety, check out tools, report locations, and log maintenance incidents directly from their cellular phones.

1. Instant Status Queries & Specification Lookup

Engineers can query any tool in the company inventory by transmitting its serial code or asset ID to the WhatsApp Bot. The bot instantly responds with the asset's current availability status, calibration certificate validity, assigned zone, and technical specs.

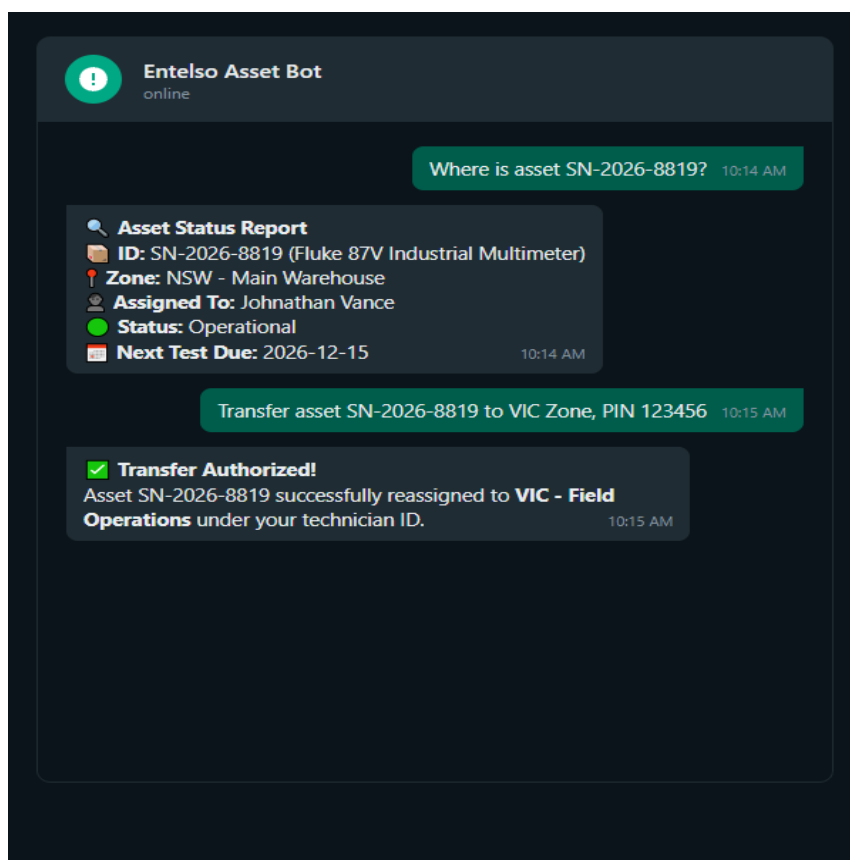


Figure 2.1: Field engineer querying tool availability and specifications via WhatsApp Bot.

2. WhatsApp Bot Command Reference Cheatsheet

All interactions use standardized syntax commands or physical QR/NFC tag scanning. The official command reference is detailed below:

Command / Action	Syntax Example	Operational Function & System Response
Status Query	#STATUS ENT-EL-004	Returns availability status, calibration expiry date, assigned zone, and specifications.
Asset Checkout	#CHECKOUT ENT-RF-001 849201	Assigns custody to engineer, logs timestamp, and updates dashboard to In Terreno .
Custody Return	#RETURN ENT-RF-001 849201	Releases custody, marks tool available in warehouse, and prompts condition check.
Zone Transfer	#TRANSFER ENT-004 ZONA-NORTH 849201	Transfers custody directly between regional zones or field sites without returning to base.
QR / NFC Scan	Scan physical tag	Automatically inputs serial code into chat and displays interactive checkout/return buttons.
Maintenance Report	#REPORT ENT-004 Screen cracked	Opens maintenance ticket, alerts supervisor on web dashboard, and requests photo.

3. Security PIN Authentication & Geolocation Tagging

Every checkout, transfer, or return command requires 2-factor authentication combining the engineer's registered cellular phone number and their secure **6-Digit Security PIN**. When executing field transfers, the bot prompts engineers to share their WhatsApp GPS location to verify physical tool presence at the job site.

4. Maintenance Incident Logging & Photo Attachments

When field engineers encounter damaged equipment or calibration drift, they initiate an immediate report via **#REPORT**. Attaching photographs directly in the WhatsApp chat uploads the images to Supabase and embeds them into the web dashboard audit drawer.

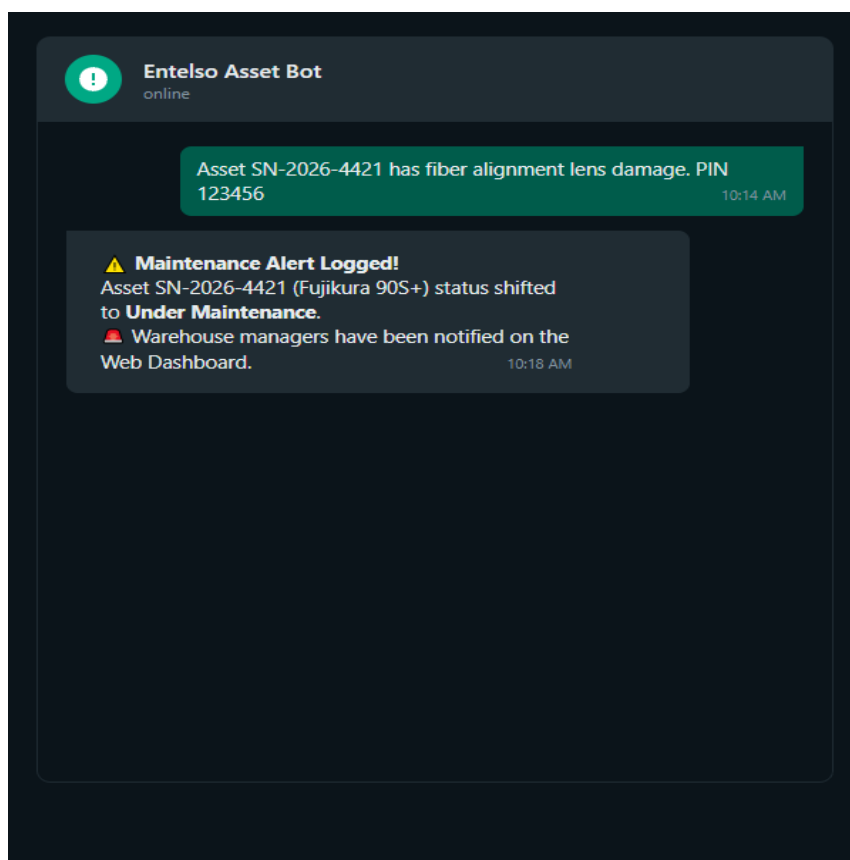


Figure 2.2: Engineer logging a field maintenance ticket with photographic evidence via WhatsApp Bot.